

MICHAEL GARCIA

Data Engineer | PhD in Physics | Modern Data Stack Expert

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ABOUT ME

Physics PhD & Data Engineer specializing in the Modern Data Stack (dbt, Cloud, CI/CD). I translate complex challenges into reliable products that create immediate business value.

EDUCATION

- 2026 **Data Engineering Zoomcamp (200h Training)** – [DataTalks.Club](#)
Advanced specialization: Batch & streaming pipeline architecture. Mastery of Cloud ecosystem (AWS/GCP), dbt, Snowflake, Spark, Kafka, and Terraform (IaC). Orchestration with Kestra.
- 2024 **Data Engineering Intensive Training (400h Training)** – [Data-Upskilling](#)
Software engineering applied to Data: Python (production-ready), SQL expert, Orchestration (Airflow), Git, and CI/CD workflows.
- 2013 **PhD in Physics for Life Sciences** – [Université Grenoble Alpes](#)

TECHNICAL SKILLS

Data Architecture & Platforms

- Data Warehousing: Snowflake, BigQuery, PostgreSQL, DuckDB
- Transformation & Modeling: dbt Core, SQL expert
- Orchestration & Ingestion: Airflow, Kestra, dlt

Cloud & Software Engineering

- Infrastructure: GCP, AWS, Azure, Terraform, Docker
- Development: Python (Pandas, Polars, PySpark, Kafka), Scraping, TypeScript
- DevOps & Quality: CI/CD (GitHub Actions), Unit Testing (Pytest), Monitoring (Grafana)

PROFESSIONAL EXPERIENCE

- 2026 **Data Engineer Consultant (Freelance)** – [Wetall \(E-commerce & Affiliation\)](#)
Revenue protection through automation
- 📊 *Audit & Consulting* – Analysis of conversion losses and deployment of an automated product availability verification pipeline.
 - 🔗 *Data Ingestion* – Design of a scraping and API integration system to synchronize partner catalogs.
 - 🔧 *Stack* : Python, Web Scraping, API Integration, GitHub Actions, Automation
 - 🔗 [Wetall Website](#)
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- 2026 **Data Engineer & Analyst** – [Energy Risk for SME Project](#)
Modeling SME exposure to electricity price volatility
- 📊 *Business Impact* – P&L impact simulation (e.g., -38% gross margin) and visualization via Looker Studio.
 - 🏗️ *FinOps Architecture* – Ingestion pipeline to BigQuery. Cost and performance optimization via partitioning and clustering.
 - 🔧 *dbt Pipeline* – OPSD data transformation: Unpivot and dynamic Cross Join with business segments (Bakery, Retail).

 *Stack : GCP, BigQuery, dbt Core, Terraform, Python, Looker Studio*

 [GitHub Code](#) · [Looker Dashboard](#)

2026

Data Engineer (Alpine Lambda Trade Project) – [Market Intelligence](#)

Real-time analysis and volatility alerts (BTC/ETH)

 *Hybrid Architecture* – Design of a Lambda architecture (Batch + Streaming) for processing live Bitcoin and Ethereum prices on Snowflake.

 *Ingestion & Monitoring* – Bollinger Bands analysis to detect rapid variations and trigger automatic alerts (sharp surges/drops).

 *Stack : Snowflake, AWS, Kafka, Python, Terraform, dbt, Kestra*

 [GitHub Code](#)

2026

Cloud Data Engineer (Grade-Scale Project) – [AI Evaluation Engine & Cloud Infrastructure](#)

Automated Azure Cloud architecture and AI agents

 *Infrastructure as Code* – Automated deployment on Azure via Terraform to ensure environment reproducibility.

 *Backend & AI* – Design of a scalable backend integrating intelligent analysis via LLMs (Groq).

 *Stack : Terraform, Azure, NodeJS, TypeScript, Zod, Groq LLM*

 [GitHub Code](#) · [Live Demo](#)

2013 – 2026

Teacher / Software Developer – [French Ministry of National Education](#)

Project Management & Pedagogy

 *Translation* – Needs analysis and automation of administrative processes using custom Python tools.

 *Knowledge Transfer* – Simplification of complex technical concepts for non-expert audiences.

2009 – 2013

PhD Researcher – [LIPhy - CNRS Grenoble](#)

Data Engineering & Modeling

 *ETL Pipelines* – Automation of the imaging pipeline, reducing analysis time by 50%.

 *Data Quality* – Algorithms for detection, reconstruction, and cleaning of trajectories (Bronze/Silver/Gold).

 *Stack : Python, IDL (matlab-like), Image Processing, Data Modeling*

LANGUAGES

French:

Native

English:

Professional (C1 - Fluent)